TUNDUMA TOWN COUNCIL
SUMAGHA SECONDARY SCHOOL
FORM FOUR MOCK EXAMINATION

142/1 INFORMATION AND COMPUTER STUDIES
PAPER 1 — THEORY

Time: 3 Hours

Instructions

1. This paper consists of sections A, B and C with a total of ten (10) questions.
2. Answer ALL questions in section A, THREE (3) questions from section B, and TWO (2) questions from section C.
3. Section A carries 20 marks, Section B carries 45 marks, and Section C carries 35 marks.
4. All questions in a section carry equal marks unless otherwise stated.
5. Cellular phones and any unauthorized materials are not allowed in the examination room.
6. Write your Examination Number at the top of every page of your answer booklet(s).

SECTION A: (20 Marks) — Answer ALL questions in this section

1. For each of the items (i) – (x), choose the correct answer from the given alternatives and write its letter in the space provided. *(10 marks)*

(i) A device that converts digital signals into analogue signals and vice versa is called a: A) Router B) Modem C) Switch D) Hub

(ii) The brain of the computer is the: A) RAM B) Hard disk C) CPU D) Motherboard

(iii) Which of the following is an example of an output device? A) Scanner B) Keyboard C) Projector D) Mouse

(iv) HTML stands for: A) Hyper Text Markup Language B) High Text Made Language C) Hyper Transfer Markup Language D) Home Tool Markup Language

(v) A network confined to a small geographical area such as a school is called a: A) WAN B) MAN C) LAN D) PAN

(vi) Unauthorized access to a computer system with intent to cause harm is known as: A) Hacking B) Booting C) Formatting D) Debugging

(vii) Which software is used to manage and organize data in tables, records and fields? A) Word processor B) Spreadsheet C) DBMS D) Presentation software

(viii) In a spreadsheet, the intersection of a row and a column is called a: A) Field B) Cell C) Table D) Range

(ix) The smallest unit of data represented in a computer is a: A) Byte B) Bit C) Nibble D) Word

(x) The stage in the System Development Life Cycle where user requirements are gathered is called: A) Design B) Testing C) Analysis D) Implementation

2. Match the item in Column A with the correct response in Column B by writing the letter of the correct response against the item number. *(10 marks)*

COLUMN A	**COLUMN B**
(i) Virus	A. A set of rules governing data exchange between computers
(ii) Protocol	B. Permanent storage that retains data when

	power is off
(iii) Firewall	C. A malicious program that replicates and spreads itself
(iv) ROM	D. A security system that monitors and controls network traffic
(v) Primary key	E. A field that uniquely identifies each record in a table
	F. Read-only memory that stores the boot-up instructions

SECTION B: (45 Marks) — Answer THREE (3) questions from this section

3. (a) Define the term computer network. (2 marks)

(b) Distinguish between LAN, MAN and WAN, giving one example of each. (9 marks)

(c) State four (4) advantages of networking computers in a school laboratory. (4 marks)

4. (a) Explain the term database management system (DBMS). (2 marks)

(b) Describe four (4) advantages of storing data in a database compared to a manual filing system. (8 marks)

(c) Explain the meaning of the following database terms: (i) Table (ii) Field (iii) Record (iv) Primary key. (5 marks)

5. (a) Explain the term Systems Development Life Cycle (SDLC). (2 marks)

(b) List and briefly explain four (4) stages of the SDLC. (8 marks)

(c) Give three (3) reasons why feasibility study is important before developing a new system. (5 marks)

6. (a) Define the following terms as used in web design: (i) HTML (ii) Tag (iii) Attribute (iv) Hyperlink. (6 marks)

(b) Write HTML code to create a simple web page with a heading 'My School' and one paragraph describing your school. (6 marks)

(c) State three (3) differences between a website and a webpage. (3 marks)

7. (a) Explain the meaning of computer ethics. (3 marks)

(b) Discuss five (5) computer crimes that affect the use of ICT in society today. (10 marks)

(c) Suggest two (2) measures a school can take to protect its computer laboratory from cyber threats. (2 marks)

SECTION C: (35 Marks) — Answer TWO (2) questions from this section

8. "The use of ICT in Tanzanian schools has both positive and negative impacts on students." With relevant examples, discuss this statement. *(17.5 marks)*

9. Explain the functions of the following computer hardware components, and for each state one common fault that may occur and how it can be fixed: (a) Motherboard (b) Power supply unit (c) Hard disk drive (d) RAM (e) Cooling fan. *(17.5 marks)*

10. You have been asked to design a simple student-registration database for your school. (a) Identify at least four (4) tables you would create and two (2) fields for each table. (b) Explain how you would ensure data integrity in this database. *(17.5 marks)*

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142/2 INFORMATION AND COMPUTER STUDIES
PAPER 2 — PRACTICAL
Time: 2 Hours 30 Minutes

Instructions

1. This paper consists of THREE (3) questions. Answer ALL questions.
2. Question 1 carries 30 marks, Question 2 carries 35 marks, and Question 3 carries 35 marks.
3. Save all your work in a folder named with your Examination Number on the Desktop before printing.
4. Print all your completed work and attach it to your practical answer sheet.
5. Raise your hand if you experience any technical/hardware problem — do not attempt to fix it yourself.
6. Switch off the computer and tidy your workstation at the end of the examination.

QUESTION 1 — WORD PROCESSING (30 Marks)

Using a word processor (e.g. Microsoft Word), do the following:

(a) Create a new document and type a one-page letter (about 150 words) inviting parents to a Form Four parents' meeting at Sumagha Secondary School. (10 marks)

(b) Format the letter as follows: (i) Title in bold, size 16, centered (ii) Body text justified, font size 12 (iii) Insert a header with the school name and a footer with page number. (8 marks)

(c) Insert a table of 4 rows and 3 columns showing a sample class timetable for Monday, with shading on the header row. (7 marks)

(d) Save the document as WORD1_[your exam number] and print one copy. (5 marks)

QUESTION 2 — SPREADSHEET (35 Marks)

Using a spreadsheet package (e.g. Microsoft Excel), create a workbook to record marks for six (6) students in three (3) subjects (Mathematics, ICS, English) as follows:

(a) Enter the students' names and marks (out of 100) for each subject. (8 marks)

(b) Use formulas/functions to calculate for each student: (i) Total marks (ii) Average marks (iii) Grade, using the criteria: A = 75–100, B = 65–74, C = 50–64, D = 30–49, F = below 30 (use the IF function). (15 marks)

(c) Use functions to calculate the class average, the highest mark and the lowest mark for each subject. (7 marks)

(d) Create a column chart comparing the total marks of all students, with a chart title and axis labels. Save the file as EXCEL1_[your exam number]. (5 marks)

QUESTION 3 — DATABASE AND WEB DESIGN (35 Marks)

PART A: Database (20 marks) — Using a DBMS package (e.g. Microsoft Access):

(a) Create a database named SCHOOL_[your exam number]. (2 marks)

(b) Create a table named STUDENTS with the fields: AdmissionNo (Primary Key), FullName, Form, Gender, Age. Set appropriate data types for each field. (8 marks)

(c) Enter data for five (5) students into the table. (5 marks)

(d) Create a query that displays the names of students who are in Form Four only. (3 marks)

(e) Create a simple form for entering new student records. (2 marks)

PART B: Web Design (15 marks) — Using a text/HTML editor:

(f) Create an HTML page titled 'Sumagha Secondary School' with a level-1 heading showing the school name. (4 marks)

(g) Add a paragraph describing the school and an unordered list of three (3) subjects offered. (5 marks)

(h) Insert a hyperlink to the Tanzania NECTA website (https://www.necta.go.tz). (3 marks)

(i) Save the file as WEB1_[your exam number].html and open it in a browser to confirm it displays correctly. (3 marks)

— END OF PRACTICAL EXAMINATION —